

A High Performance Approach for Classification of Motor Imagery EEG

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Abstract—Use of Motor Imagery in EEG signals is gaining importance to develop Brain Computer Interface (BCI) applications in various fields ranging from bio-medical to entertainment and its classification at lower complexity in EEG signals is gaining importance. Filter Bank Common Spatial Pattern (FBCSP) has been a promising feature extraction approach to deal with subject-specific behavior in Motor Imagery classification. Though its computational complexity has been on the lower side, the approach has not yet matched the classification accuracy of non-FBCSP approaches. By introducing a combined feature set of Band Power (BP) features and Time Domain (TDP) features in FBCSP approach we have succeeded in proposing an approach of high classification accuracy while out competing the computational complexity of the best reported non-FBCSP approaches. We have also analyzed the impact of parameter variations on classification accuracy and achieved 0.59 mean kappa value for Dataset 2a BCI competition IV on four class problem. This is the highest reported for FBCSP approaches along with the lowest inter-subject variation.

Index Terms—Motor Imagery, Brain Computer Interface, Electroencephalography

I. INTRODUCTION

Brain computer interface (BCI) technology represents an immensely growing field of research with application systems. Motor Imagery (MI) Electroencephalography (EEG) based Brain Computer Interfaces have numerous real-time applications like wheelchair or prosthetic devices for disabled people and in gaming. The BCI interface for this purpose should have high MI task classification accuracy, low computational complexity, low response time and low intra-subject and inter-subject variability. There is also an emphasis on less number of channels and adaptability to new subjects. A practical high performance approach should therefore offer a good compromise between these criteria if not the best. For example high accuracy but at the cost of high latency will not be acceptable.

Most of the two class MI approaches are based on Common Spatial Pattern (CSP) filtering of the EEG signal. CSP can be used to discriminate two classes based on the maximization of variance of particular classes [1]. For four-class problem, variant of CSP approach has been used. Local Temporal Common Spatial Pattern (LTCSP) [2], utilize the time-dependent neighboring matrix for maximization of co-variance. Linear Discriminant Analysis (LDA) classifier is used in One-versus-rest (OVR) configuration and they obtained 0.61 mkv (mean kappa value) for four class EEG data. Spatial-frequency-temporal features were proposed in [3]. Fisher Ratio algorithm

was used for the selection of subset channels and then they decomposed the signals into time-frequency components. Sparse regularization and CSP was applied for feature extraction. They obtained 0.53 mean kappa value using Naive Bayesian classifier.

Filter Bank Common Spatial Pattern (FBCSP) [4] was introduced to optimize subject specific frequency bands for CSP. It has been effectively extended for use of four class problem. Researchers tried different classification methods with FBCSP approach extending it for multi-class data i.e. Pair-Wise (PW), Divide and Conquer (DC), One-Versus-Rest (OVR) and Naive Bayesian Parzen Window (NBPW) and achieved 0.57 mean kappa value by using FBCSP with OVE and PW classification method.

As selected feature for classification, Average Band power of signal has been mostly used [5] [12] which enables lowering of dimensionality of data. In the literature there is also a considerable number of publications which has used Time Domain Parameters (TDP) to mine temporal features. These features have been first proposed and shown effective in 2-class problems in [13] and [14]. In [6], TDP is used in terms of activity, mobility, and complexity to compute velocity and other features in extension and flexion positions of an index finger, middle finger and thumb.

In this paper we propose a high performance approach based on FBCSP projection as this has been shown to have reasonable complexity and provide a discriminative decomposition for multi-class classification problem. The novelty of our approach is in presenting a new combination of features and our extensive experimentation to find optimal parameters. We are proposing a feature set as fusion of TDP and BP features. Many experts regard subject specific time segment and frequency band selection as still a challenging task. In this study, we have worked on optimizing the appropriate time segments, frequency bands, smoothing window for BP, the order of derivative for TDP and feature window. This has yielded an approach which has high accuracy, low computational complexity and minimal response time. It also shows less inter-subject variation as well.

II. PROPOSED METHOD

The proposed approach for multiclass motor imagery classification is shown in Fig. 1. This approach is comprised of sub blocks: preprocessing, band pass filtration, common

spatial pattern, feature computation and classification. In preprocessing stage, artifacts are reduced and then signal is divided into subbands and band pass filters are applied on each individual band. In CSP stage, signals are mapped in spatial domain to increase dissimilarity among the features. In feature computation stage, TDP and BP features are computed and combined to form final feature vector which are fed to the classifier for classification. Additional details and background of the proposed algorithm can be found in [11] [7].

A. Preprocessing

Artifacts reduction, frequency bands (μ, β) selection and windowing operations are performed in preprocessing stage. Artifacts can decrease the performance of EEG classification which are must to be removed before the computation of features. Event related phenomena (Event Related Synchronization and Event Related Desynchronization) are the features in which we are interested. These events occur in μ (8–13 Hz) and β (13–30 Hz) bands. To achieve these goals, we used butterworth band-pass filter of order 5th. Each single trial was conducted for 8.0 sec and subjects imagined the execution of motor imagery task between 3.0–6.0 sec. Actual motor imagery phenomena can lie in different duration for each individual, so we considered time segmentation parameter in this study and performed time segmentation of 3.5–6.0 sec and 4.0–6.0 sec on average for BP and TDP features respectively in the testing phase. These time intervals were determined in the training phase.

B. Feature Extraction

For feature extraction, Filter Bank Common Spatial Pattern (FBCSP)[4] is used to maximize dissimilarity between all classes which contributes towards the final feature set. Progressive stages of FBCSP combined with other stages

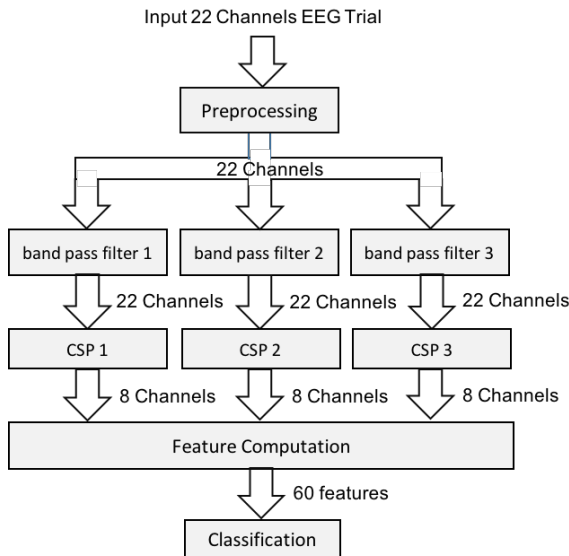


Fig. 1: Proposed Model for classification of multi-class motor imagery EEG.

of the proposed approach are shown in Fig. 1. It consist of three progressive stages: Bandpass filtration for multiple bands, spatial filtration using Common Spatial Pattern (CSP) algorithm and feature selection.

C. Feature Computation

Discriminant features, obtained form FBCSP approach, are used to compute Band Power (BP) and Time Domain Parameters (TDP) which compute spatial and temporal features respectively. Utilization of TDP with BP is the novelty of the proposed approach as only BP features are not enough to classify multiclass EEG data, comparatively. Temporal information fused with spatial features could discriminate EEG classes in a better way which is the motivation of combining BP and TDP features in this research.

1) *Band Power (BP)*: Features form intentness and average power of EEG trials can be computed using Band Power (BP) where difference between instantaneous and average power is divided by the average power which normalize each individual trial [10] BP feature can be obtained using following equations:

$$\vec{P}(t) = \vec{V}^2(t) \quad (1)$$

Where $\vec{P}$ and $\vec{V}$ are the time course of power and the input respectively. Smoothing window is used to smooth the BP features. Hence onset and exact motor imagery action vary person to person so empirical smoothing window size is used. Average power $P_{avg}(\tau)$ can be computed as:

$$P_{avg}(\tau) = \frac{1}{M} \sum_{i=1}^M \vec{P}(t_i + \tau) \quad (2)$$

where M is number of trials. τ is the time of a trial. Event-Related Desynchronization (ERD) is short-lasting attenuation in μ and β bands which measures absolute mean of the ratio between power difference (instantaneous and average) and average power as:

$$\mathbf{F}^{BP} = \text{mean} \left(\left| \frac{\vec{P}(t) - P_{avg}(\tau)}{P_{avg}(\tau)} \right| \right) \quad (3)$$

Where $\mathbf{F}^{BP}$ is ERD feature set obtained using instantaneous power ($\vec{P}(t)$) and average power (P_{avg}) which is calculated against per frequency band per time segment. We used smoothing window of size 2.0 sec to save computational cost.

2) *Time Domain Parameters (TDP)*: TDP is another feature which is used in contrast to BP to exploit maximum required information hidden in EEG trials. To obtain time domain parameters, the variance of power is calculated with the first near-optimal k number of derivatives as:

$$\vec{P}_i(t) = \left(\frac{d^i \vec{V}(t)}{dt^i} \right) \quad (4)$$

Where $\vec{P}_i[n]$ is the sample obtained form $\vec{P}_i(t)$ and $\vec{P}_i(t)$ is i^{th} derivative of input signal $\vec{V}(t)$. IIR filter is applied on $\vec{P}_i(t)$ with an exponential moving average window to smooth

the signals. Natural log of the variance of resulting output is computed to normalize the features.

We obtained 24 features from BP and 36 features from TDP and final feature set contains 60 (24 + 36) features which are finalized based on training the system. The 36 most effective TDP features were selected out of possible 120 by looking at the system performance during cross validation. The simplified training procedure was based on dividing the entire set into smaller subsets and choosing the best performing subset of TDP features.

D. Classification

Linear Discriminant Analysis (LDA) and Support Vector Machine (SVM) are used separately for training and testing of the proposed model. Ground truth is labeled in numbers (1, 2, 3, 4) for four classes of MI data. Hence both classifiers are binary classifiers, to classify multi-class data, LDA classifier is used with OVR configuration and Gaussian kernel is used for SVM.

III. RESULTS

Before proceeding to discuss our experimental results we discuss the data set used and the evaluation matrix.

A. Dataset and Evaluation Metrics

We used BCI Competition IV [9] Dataset 2a comprised of four classes (left, right, feet and tongue) of motor imagery measurements collected from 9 subjects. Dataset recorded in 2 sessions, one for training and one for testing and evaluation. There are total 288 trials recorded using 22 EEG channels and 3 EOG channels.

Cohen's Kappa (κ) is used as evaluation matrix. It is recommended for evaluation by BCI competition IV committee. κ is a robust statistics used for testing of reliability of intrarater or interrater. $\kappa = 0$ represent the number of agreements occurred by chance (randomly) and $\kappa = 1$ represent full agreement whereas $\kappa \leq 0$ results if agreement becomes less than the agreement by chance.

B. Parameter Optimization

Using multiple experiments, we selected the optimal parameters: frequency bands (8 – 14 Hz and 20 – 16 Hz for BP and 8 – 15 Hz and 19 – 24 Hz for TDP), Butterworth filter, smoothing window of size 2.0 s, upto 5th orders of TDP features, kernel width and soft margin of value 100 for SVM classifier. We got best results with using SVM classifier which is the motivation of selection of SVM classifier in the proposed model.

The results of the proposed method with the optimized parameters are shown subject-wise in Table I in the second column. Our proposed approach has achieved the mean kappa value of 0.59 and the standard deviation of 0.124.

IV. DISCUSSION ON PERFORMANCE

In this section we like to compare our results with the state-of-the-art results and like to highlight how performance of our proposed approach provides a better compromise compared to existing one between accuracy, inter-subject stability, computational complexity and response time.

A. Accuracy and Inter-subject Stability

Table I provides a comparison of our approach with FBCSP approaches reported in [8] and also other competitive approaches like [9], [4], [3], [2]. We see that our approach which is also FBCSP based clearly outperforms the approaches in [8]. In [9], researchers were able to achieve 0.58 MKV where they obtained standard deviation of 0.268. As compared to FBCSP approaches, proposed features (BP and TDP) combined with FBCSP, outperformed the reported methods in literature with both LDA and SVM classifiers [9] [4] [3] [8] by yielding 0.59 MKV and 0.124 standard deviation as shown in Table I. In [2], researchers used LTCSP and achieved 0.61 MKV but still have a greater standard deviation of 0.196. Though we have not tried LTCSP yet, even with FBCSP the BP and TDP features have resulted in a comparable accuracy (0.58 mkv) and show less variation in performance over a different subject.

B. Computational Complexity

In testing, a single input to the classifier can be formulated as $R^{n \times d}$, where R is input sample of dimension $n \times d$. In [4] researchers used FBSCP algorithm for feature extraction and Parzan Window (PZ) for classification of multiclass MI data. Computational complexity of FBCSP algorithm is $O(m3 + Rm2)$, where R is sample data element and m is the number of features extracted using CSP algorithm. PZ classifier has computational complexity $O(n, d, R)$ and overall complexity of the method proposed in [4] can be written as: $O(m3 + Rm2) + O(n, d, R)$.

In [9] Covariate Shift Detection (CSD) with adaptive learning capability was used and comprised of three stages: Moving average filter, Kolmogorov–Smirnov statistical hypothesis test (K–S test), and Hotelling T-Squared multivariate statistical hypothesis test (HT-S test). The computational complexity of K-S test and HT-S test can be formulated as $O(n \log n)$ and $O(n)$. They used SVM classifier for the classification of multiclass data. Overall complexity of their approach can be defined as: $O(Nn) + O(n \log n) + O(n) + O(nd)$.

LTCSP method [2] does not consider the inter-class variance and it requires manual selection of weight parameters which is more computationally expensive as compared to FBCSP. They used LDA classifier in OVR mode. Overall computational complexity of this approach can be formulated as: $O(m3 + Rm2) + O(nd^2)$. In contrast, we used FBCSP for feature extraction and SVM classifier. Overall testing computational complexity of our proposed model can be written as: $O(m3 + Rm2) + O(nd)$. It can be seen that the proposed model has less computational complexity as compared to compared methods.

C. Training Performance

Feature set is another important factor of complexity as large feature set takes more time to compute and to train the classifier. In [9], researcher used 220 features to train and test the classifier. Best reported method [2] in literature used 72 features as final feature set. In contrast, we used feature set of

TABLE I: Classification accuracy comparison (Kappa Coefficients) on BCI Competition IV dataset 2a, Key: MKV: Mean Kappa Value

Subjects	Proposed method (FBCSP)	Raza, H. [9]	Ang, Kai Keng[4] DC	OVR	PW	Miao, Minmin [3]	Ghabri, H. [2]	1st	2nd	3rd	4th	5th
1	0.68	0.88	0.71	0.68	0.78	0.63	0.74	0.68	0.69	0.38	0.46	0.41
2	0.43	0.22	0.37	0.42	0.40	0.43	0.43	0.42	0.34	0.18	0.25	0.17
3	0.70	0.88	0.66	0.75	0.75	0.74	0.80	0.75	0.71	0.48	0.65	0.39
4	0.61	0.39	0.41	0.48	0.52	0.54	0.57	0.48	0.44	0.33	0.31	0.25
5	0.80	0.53	0.40	0.40	0.41	0.19	0.31	0.40	0.16	0.07	0.12	0.06
6	0.42	0.33	0.26	0.27	0.19	0.26	0.38	0.27	0.21	0.14	0.07	0.16
7	0.55	0.38	0.73	0.77	0.80	0.63	0.76	0.77	0.66	0.29	0.00	0.34
8	0.55	0.85	0.58	0.75	0.74	0.62	0.75	0.75	0.73	0.49	0.46	0.45
9	0.56	0.81	0.50	0.61	0.54	0.69	0.80	0.61	0.69	0.44	0.42	0.37
MKV	0.59	0.58	0.52	0.57	0.57	0.53	0.61	0.57	0.52	0.31	0.30	0.29
stdv.	0.124	0.268	0.165	0.183	0.212	0.192	0.196	0.183	0.229	0.153	0.213	0.134

size 60 for training and testing of the classifier. Small feature set shows the credibility of the proposed model for real-time applications.

D. Response Time

The response time of the system in terms of interpreting the MI action once it has begin in EEG is determined by the time window used for processing and computational delay. We have favorably compared the computational complexity of our approach with others in the above paragraph. Most of the approaches we are comparing with use a time window of 2 s like [8] [3]. Our time window is 2 to 2.5 s. This is comparable with the generally reported ones. However, [2] which reports the highest kappa value, uses a time window of 3 s which is higher than ours.

V. CONCLUSIONS

The literature survey on existing approaches for four-class MI classification shows that non-FBCSP approaches have higher accuracy compared to FBCSP approaches. However, the FBCSP approaches are comparatively computationally light. We have experimented with an approach that introduces BP and TDP features in FBCSP algorithm and has succeeded in proposing an approach which has considerably increased the recognition accuracy of the FBCSP algorithm. Hence, yielding an approach more suited for real-time BCI application. Use of TDP features with FBCSP is reported for the first time as to our knowledge. The near-optimal frequency bands and time segment in a trial to extract MI feature also vary from subject to subject. Our method also investigate optimal time-frequency range for subject-specific feature extraction hence allowing lesser inter-subject variation. The proposed method was evaluated on BCI Competition IV Dataset 2a (four class MI data). Our method is robust to artifacts and stable over all 9 subjects. As future work, we will combine LTCSP with BP and TDP to extend our experiments toward an increase in classification accuracy.

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